

Crop-ranking reversal under downside risk and operational constraints

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Abstract

Crop recommendation systems rank crops using agronomic or predicted performance, whereas producers allocate land under uncertain margins, joint downside risk and operational restrictions. We ask when these two orders reverse and how diversification and climate information alter the resulting exposure. We formulate a stochastic crop-planning programme with uncertain price, yield and cost; land, budget, rotation, contract, labour and equipment constraints; and a loss-CVaR ceiling. A restricted two-crop theorem gives a necessary-and-sufficient reversal frontier: when the higher-ranked crop also has the higher expected margin, reversal occurs exactly when no risk-feasible allocation remains on the upper half of its feasible share interval. The result is family-specific and need not reduce to a universal tail-dependence threshold. A public-data-calibrated Kansas experiment uses pre-decision relative-yield potential as a genuine agronomic score. Winter wheat ranks first (0.894) but receives the least land (0.248), below soybean (0.422) and corn (0.330), in the Student- t model. Reversal appears in 143 of 165 copula-dependence-risk cells; 96 are selected complete rank reversals and 0 are strong exclusion reversals. The Kansas inversion is margin-induced and is moderated by risk. Separate controlled experiments identify a risk-induced Soybean–Corn crossing with the higher-score crop also having the higher mean margin, and an operational crossing under a tightened rotation cap. A full-investment matched-Kendall Gaussian mean-variance policy selected by a fixed 15% variance-reduction rule reduces variance relative to an expected-profit benchmark yet violates the Student- t evaluation-law CVaR ceiling. Strong failure holds over $\gamma \in [0.0068 - 0.0088]$ in the focal experiment; because the tail-aware optimum binds the ceiling, tail inferiority and ceiling violation coincide numerically. In a two-stage extension, adjacent-grid cross-differences under one shared ex-ante CVaR ceiling are positive, zero or negative depending on what flexibility changes. A separate 248-state-year public panel shows widespread score-acreage disagreement but no leakage-free lagged association, so it validates prevalence only descriptively. Ranking reversal is therefore a conditional decision-system property, not a generic defect of ranking and not evidence that observed acreage is privately optimal.

1 Introduction

Crop recommendations are often communicated as an order: crop i is more suitable than crop j , or a predictive model assigns i the larger yield potential. Planting is not an order. It is a vector of land commitments made before prices, yields and operating conditions are fully known. The distinction becomes consequential when margins co-move in bad years, when a lender caps working capital, when rotation rules restrict crop shares, when a contract requires minimum acreage, or when labour and machinery are shared across crops. A producer can then rationally plant less of the highest-ranked crop without rejecting the agronomic information contained in the ranking.

This paper studies that event as *crop-ranking reversal*. For a pair with score $s_i > s_j$, reversal means $x_i < x_j$ in the selected optimal acreage plan. A complete inversion occurs when the top-ranked crop receives less land than every lower-ranked crop. Strong reversal is narrower: a higher-ranked crop is excluded, $x_i = 0 < x_j$, while a lower-ranked crop is planted. The question is not whether a ranking alone identifies acreage. It does not. The scientific question is sharper: under a declared stochastic crop-planning model, which economic, risk and operational mechanisms move the optimum across the

ranking boundary, where is the resulting frontier, and when do conventional diversification or additional information fail to protect the decision?

Three features make this question difficult. First, the score must remain conceptually distinct from profit. A revenue score relabelled as “suitability” would make the comparison circular. We therefore construct a genuine pre-decision agronomic index from historical state yield relative to same-year national yield. Expected margin is constructed separately from price, yield and operating cost. Second, joint downside risk cannot be recovered from marginal standard deviations or a single rank-correlation coefficient. Gaussian, Student- t and Clayton copulas can match Kendall’s τ while assigning different mass to joint crop failures. Third, a solution returned by one linear-programming run may be one point on an optimal face. A reversal claim must distinguish whether reversal is possible, universal across the face, or selected by a declared deterministic rule.

We address all three features in one model. Acreage is chosen to maximize expected profit subject to a loss-CVaR ceiling and a polyhedral operational set containing land, idle land, crop bounds, budget, rotation, contractual minimums and shared labour and equipment. The finite-scenario form is a linear programme. Its Karush–Kuhn–Tucker system includes the free VaR threshold, scenario excess variables and every operational multiplier. The full system prevents a local marginal-value identity from being misread as an acreage ordering.

Our first contribution is an exact but deliberately restricted reversal characterization. In a fixed-land two-crop problem where the higher-ranked crop also has the larger expected margin, its optimal share is the right endpoint of the risk-feasible interval. Reversal occurs if and only if that endpoint is below one-half of total land. With additional monotonicity on the upper half, the condition reduces to CVaR at the equal-share allocation exceeding the risk ceiling. A unique dependence threshold exists only within a named family when continuity, strict monotonicity and crossing are verified. Without those conditions the reversal set can be disconnected. This rules out informal arguments that attempt to infer acreage from a “risk-adjusted margin gap” or from a scalar tail-dependence coefficient.

Our second contribution is a full-model experiment calibrated from official US agricultural series. Kansas corn, soybean and winter wheat provide 8 complete pre-evaluation years. The score order is winter wheat, soybean, corn, while mean real margins are \$ 138.3, \$ 229.0 and \$ 199.8 per acre, respectively. In the Student- t copula stress model, the 95% loss-CVaR ceiling binds. The top-ranked winter wheat receives share 0.248, less than corn (0.330) and soybean (0.422). Expected profit is \$ 198.5 per normalized acre and loss-CVaR is \$ - 47.0. KKT primal, stationarity and complementarity residuals are below 6×10^{-14} . Because the no-CVaR endpoint is already inverted and assigns winter wheat only 0.070, this Kansas result is margin-induced; the risk ceiling raises the wheat share and moderates the inversion. Separate controlled paths identify risk- and operationally induced Soybean–Corn crossings while holding the relevant score and mean-margin orders fixed.

The phase diagram is more informative than this one allocation. Across 165 pre-specified cells spanning three copula families, five Kendall- τ values and eleven risk tolerances, 143 cells show selected pairwise reversal, 96 show selected complete rank reversal and 0 show strong exclusion reversal. Two cells contain multiple optima, no cells are infeasible, and one Clayton path is disconnected on the pre-specified grid. Selected pairwise reversal occurs in 62 of 64 historical resamples, with exact Clopper–Pearson binomial interval [0.892, 0.996]. These quantities are uncertainty diagnostics for the structural stress test, not estimates of farm-level prevalence.

Our third contribution separates diversification diagnostics that are often conflated. We define x^0 explicitly and claim failure only when a full-investment Gaussian mean–variance policy strictly reduces variance relative to that benchmark, differs from the tail-aware allocation and has worse loss-CVaR under the Student- t evaluation law. The focal policy is selected independently of tail performance as the smallest γ that achieves a 15% Gaussian variance reduction. At matched Kendall’s $\tau = 0.25$, Gaussian advice has Student- t loss-CVaR \$ - 43.4, exceeding the ceiling \$ - 45.7; the tail-aware allocation attains \$ - 45.7. Strong failure holds over $\gamma \in [0.0068 - 0.0088]$ in the focal experiment. Because the tail-aware optimum binds the ceiling, tail inferiority and ceiling violation are numerically equivalent here rather

Ranking becomes allocation only through margins, joint risk and operations

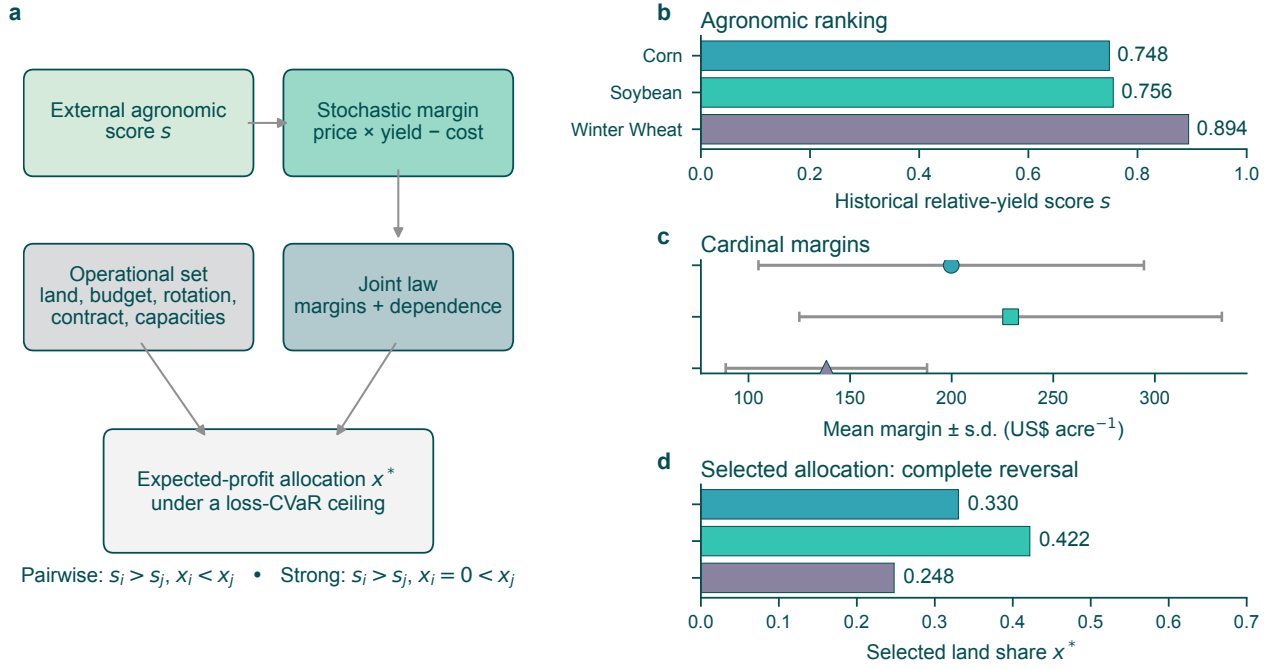


Figure 1: From agronomic ranking to risk-constrained operational allocation. **a**, The score remains external to the profit objective. Stochastic margins and their joint law enter expected profit and loss-CVaR, whereas land, budget, rotation, contract and shared-capacity restrictions define feasible action. **b**, The pre-decision score ranks winter wheat first. **c**, Cardinal mean margins are estimated separately; bars show one standard deviation across the eight calibration years. **d**, The full-model optimum assigns the least land to winter wheat and therefore produces a selected complete rank reversal, not a strong exclusion reversal.

than independent evidence.

Finally, we examine the relation between information and flexibility. The signal is an imperfect pre-plant indication of relative crop-yield conditions. In one path, flexibility expands the share that can be reallocated after observing the signal. In a second path, flexibility buffers the state-dependent yield shock through recourse interpretable as irrigation, input switching or harvest timing. There, the information value necessarily falls to zero when the predicted state is fully buffered. For the complete shared-CVaR programme, adjacent-grid cross-differences are classified numerically: 32 positive, 17 negative, 18 zero-information and 5 zero/boundary. The strict theorem is restricted to the separable or nonbinding-risk submodel.

Figure 1 summarizes the architecture. It preserves the scientific chain from suitability to stochastic operational optimality: score, uncertain margin, joint law, operational set, downside-risk constraint, reversal frontier, diversification failure and information actionability. The paper does not recast the topic as a general essay about ordinal insufficiency. Ordinality is a boundary condition; the contribution is the conditional mechanism and its verified consequences.

2 Literature review and scientific position

2.1 Risk-aware crop planning

Crop-planning models have long represented planting as an operational allocation rather than a list of recommended crops. Their variables include crop areas, sequences and timing; their constraints include

land, capital, labour, equipment, rotations, storage and contracts. Contemporary models combine these objects with crop-selection integer programmes [Filippi et al., 2017], dynamic rotation benefits [Boyabath et al., 2019], sustainable rotation systems [Benini et al., 2023], contract-farming schedules [Li et al., 2015] and spatial decision support [Pahmeyer et al., 2021]. Reviews show that climate uncertainty is handled through stochastic, robust and multiobjective approaches rather than one standardized decision model [Randall et al., 2022, 2024].

Loss-CVaR is attractive in this setting because it controls an expected shortfall and admits a linear finite-scenario representation [Rockafellar and Uryasev, 2000, 2002]. Agricultural precedent is substantial but fragmented. AitSahlia et al. [2011] combine ENSO-conditioned planting and hedging with Gaussian-copula scenarios and CVaR. Filippi et al. [2017] embed CVaR in operational crop selection. Larsen et al. [2015] and Nguyen-Huy et al. [2018] use copulas and mean-CVaR to study geographical wheat portfolios. These studies establish that crop portfolios can be optimized under downside risk. They do not ask whether an externally supplied agronomic ranking reverses, characterize the ranking boundary, or bound all optima on the objective-equivalent face.

2.2 Joint tail risk and the limits of conventional diversification

Diversification is effective when crop outcomes do not fail together in the states that matter to the decision. Linear correlation summarizes average co-movement, not the geometry of the joint lower tail. Student- t copulas retain nonzero symmetric tail dependence, while Archimedean families can concentrate dependence asymmetrically [Demarta and McNeil, 2005]. Dependence comparisons are family- and order-specific [Ansari and Rockel, 2024]; a single lower-tail coefficient does not order arbitrary portfolio losses across families.

Agricultural evidence motivates this distinction. Joint yield tails vary with land quality [Du et al., 2018], production shocks are dependent across major breadbaskets [Pflug et al., 2017], and copula estimates show that the risk reduction from crop diversification changes with crop pair, region and drought severity [Schmitt et al., 2024]. These findings do not imply that diversification generally fails. They imply that “number of crops” and variance reduction are incomplete diagnostics for a loss-CVaR-constrained portfolio. Our criterion therefore requires a conventional policy to pass a fixed variance benchmark yet fail under the Student- t evaluation law and differ from the tail-aware allocation.

2.3 Prediction, prescription and suitability

Agricultural machine learning commonly predicts yield, classifies land or ranks crop performance using weather, soil, management and remote-sensing features [Liakos et al., 2018, van Klompenburg et al., 2020]. A score can be scientifically useful even when it is not an acreage prescription. The prescriptive-analytics literature instead evaluates how conditional predictions change an optimized decision [Bertsimas and Kallus, 2020, Elmachoub and Grigas, 2022]. Decision-focused learning extends this principle to end-to-end systems [Mandi et al., 2023].

Our contribution is not a new learning algorithm. We use this literature to make the interface explicit. A genuine agronomic score ranks historical relative-yield potential and never enters the profit objective. The optimizer then reveals whether the operationally optimal area order agrees with that score. This design avoids two errors: treating prediction accuracy as proof of prescriptive optimality, and dismissing the score because it does not contain every economic variable.

2.4 Information value and operational flexibility

Seasonal climate information has value only through decisions it can change. Reviews emphasize that forecast skill, risk preferences, institutions and available management responses jointly determine economic value [Meza et al., 2008]. Bioeconomic applications measure value through changes in sowing and other decisions [An-Vo et al., 2021], while ENSO-based crop-planning models combine forecasts with planting

Table 1: Position relative to the closest research streams. A check indicates a central analyzed object rather than a passing mention.

Stream	Agro. rank	Operations	CVaR	Tail law	Frontier	Info-flex
Agricultural ML reviews	✓					
Crop-planning optimization		✓	some	limited		
Agricultural copula-CVaR		partial	✓	✓		
Climate forecast valuation		partial	some			partial
This study	✓	✓	✓	✓	✓	✓

and financial hedges [AitSahlia et al., 2011]. General decision analysis likewise links information value to the portfolio or action set [Keisler, 2004, Merkhofer, 1975]; operational-flexibility theory shows that flexibility value depends on margins, cost and the full uncertainty structure [Van Mieghem, 1998].

None of this supports unconditional strict complementarity. A flexible action can make a signal actionable, but it can also provide a robust fallback that reduces the need to distinguish states. We therefore prove non-negativity, strictness and zero conditions separately, add an increasing-differences condition for complementarity, and construct an agricultural buffering path for substitution.

2.5 Contribution matrix

The novelty is thus a conjunction, not a claim that no previous study has used any individual ingredient. We connect a genuine agronomic rank to a complete stochastic operational model; give a valid reversal frontier; separate variance, dependence and CVaR notions of diversification; and determine when information complements, substitutes for or is unrelated to flexibility.

3 Stochastic crop-planning model

3.1 Scores, margins and joint uncertainty

Let $i = 1, \dots, n$ index crops and x_i denote planted area. A pre-decision score s_i induces a crop order. It is external to the optimization model: no cardinal transformation of s is used as a margin. For state ω , stochastic per-acre margin and portfolio profit are

$$\tilde{\pi}_i(\omega) = \tilde{P}_i(\omega)\tilde{Y}_i(\omega) - \tilde{C}_i(\omega), \quad \tilde{\Pi}(x, \omega) = \sum_i x_i \tilde{\pi}_i(\omega). \quad (1)$$

The joint distribution includes within-crop price–yield dependence and cross-crop co-movement. Let $\mu_i = \mathbb{E}[\tilde{\pi}_i]$ and define loss $L(x, \omega) = -\tilde{\Pi}(x, \omega)$. A larger loss-CVaR is worse; because profitable portfolios can have negative loss-CVaR, “more negative” means safer under this convention.

3.2 Operational feasibility

The operational set is

$$\mathcal{X} = \left\{ x \in \mathbb{R}_+^n : \mathbf{1}^\top x \leq A, \quad c^\top x \leq B, \quad Rx \leq r, \quad Kx \geq k, \quad Gx \leq h, \quad \ell \leq x \leq u \right\}. \quad (2)$$

Here A is available land, c and B describe pre-plant cash requirements, R contains rotation caps, K contains contract minimums, and G contains shared labour, equipment, irrigation or storage rows. Idle land is $A - \mathbf{1}^\top x$ and is reported even when it is not assigned a separate variable. These constraints are not post-hoc explanations of an observed acreage vector; they define the controlled decision problem.

3.3 Loss-CVaR ceiling

For confidence level α ,

$$r_\alpha(x) = \text{CVaR}_\alpha[L(x)] = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1-\alpha} \mathbb{E}[(L(x) - v)_+] \right\}. \quad (3)$$

The producer solves

$$\max_{x \in \mathcal{X}} \mu^\top x \quad \text{subject to} \quad r_\alpha(x) \leq \kappa. \quad (4)$$

For scenarios $m = 1, \dots, M$ with probabilities w_m , free threshold v and excess variables $q_m \geq 0$, the risk restriction is

$$v + \frac{1}{1-\alpha} \sum_m w_m q_m \leq \kappa, \quad (5)$$

$$q_m \geq -\pi_m^\top x - v. \quad (6)$$

The resulting programme is linear. The risk ceiling is indexed by $\rho \in [0, 1]$:

$$\kappa(\rho) = \kappa_{\min} + \rho(\kappa_{\text{EP}} - \kappa_{\min}), \quad (7)$$

where $\kappa_{\min}$ is minimum attainable loss-CVaR and κ_{EP} is the loss-CVaR of the unconstrained expected-profit endpoint. This makes risk tolerance comparable across joint laws without transferring an arbitrary dollar ceiling.

3.4 Reversal and optimal faces

Definition 1 (Pairwise, complete-rank and strong reversal). *For a pair $s_i > s_j$, an allocation reverses the ranking if $x_i < x_j$. A complete rank reversal, or top-crop rank inversion, occurs when the highest-ranked crop has less area than every lower-ranked crop. Strong reversal requires $s_i > s_j$ and $x_i = 0 < x_j$.*

Let

$$\mathcal{S}_\kappa = \arg \max \{ \mu^\top x : x \in \mathcal{X}, r_\alpha(x) \leq \kappa \}$$

be the complete optimal set. Reversal is *possible* if it occurs at some $x \in \mathcal{S}_\kappa$, *universal* if it occurs at every such point, and *selected* if it occurs at the point returned by the deterministic selector. We examine a pair by minimizing and maximizing $x_j - x_i$ on the objective-equivalent face. A single solver vertex is not sufficient evidence of universal reversal. Ordering and operational zero use numerical tolerances of 10^{-4} normalized land share; sensitivity spans 10^{-8} to 10^{-2} .

3.5 Full stationarity system

Let nonnegative multipliers $\eta, \gamma, \beta, \rho, \chi, \lambda, a, b$ correspond to loss-CVaR, land, budget, rotation, sign-converted contract, shared capacity, lower bounds and upper bounds. For a CVaR subgradient $d \in \partial r_\alpha(x^*)$, acreage stationarity is

$$-\mu + \eta d + \gamma \mathbf{1} + \beta c + R^\top \rho - K^\top \chi + G^\top \lambda - a + b = 0. \quad (8)$$

Pairwise subtraction cancels the common land multiplier but retains budget, shared-resource and crop-bound pressures. Equation 8 is a marginal balance, not an acreage-ordering formula. The finite-scenario implementation also verifies stationarity for v and every q_m , including fractional tail weights when the loss distribution has atoms.

4 Theoretical results

4.1 An exact two-crop reversal frontier

Consider two crops with $s_1 > s_2$, fixed planted land $x_1 + x_2 = A$, and an operationally feasible interval $I = [\underline{z}, \bar{z}]$ for $z = x_1$. Under a declared joint-law path θ , write

$$r_\theta(z) = \text{CVaR}_\alpha[-z\tilde{\pi}_1 - (A - z)\tilde{\pi}_2], \quad \mathcal{K}_{\theta,\kappa} = \{z \in I : r_\theta(z) \leq \kappa\}.$$

Theorem 1 (Necessary-and-sufficient restricted reversal). *Suppose $\mu_1 > \mu_2$ and $\mathcal{K}_{\theta,\kappa}$ is nonempty. Then the unique optimum is*

$$z^*(\theta, \kappa) = \sup \mathcal{K}_{\theta,\kappa}.$$

Consequently,

$$x_1^* < x_2^* \iff \sup \mathcal{K}_{\theta,\kappa} < A/2 \iff \mathcal{K}_{\theta,\kappa} \cap [A/2, \bar{z}] = \emptyset.$$

If r_θ is nondecreasing on $[A/2, \bar{z}]$ and a risk-feasible point exists below $A/2$, reversal is equivalent to $r_\theta(A/2) > \kappa$.

Proof. The objective is $\mu_2 A + (\mu_1 - \mu_2)z$, which is strictly increasing in z . Loss-CVaR is convex and continuous in z , so its sublevel set intersected with compact I is compact. The strictly increasing objective is uniquely maximized at the right endpoint. The three reversal statements then follow from $x_1 < x_2 \iff z < A/2$. Under upper-half monotonicity, the lowest risk on that half is attained at $A/2$, giving the last equivalence. $\square$

The theorem is narrow by design. It characterizes acreage directly while retaining expected-margin ordering, operational bounds and the full joint loss law. If $r_\theta(A/2)$ is continuous and strictly increasing along a named family and crosses κ , there is a unique family-specific threshold θ^* solving $r_{\theta^*}(A/2) = \kappa$. Without strict monotonicity or crossing, the scientific object is the reversal set $\{\theta : \sup \mathcal{K}_{\theta,\kappa} < A/2\}$; it may contain several components. This is why a universal “higher tail dependence causes reversal” statement is not valid.

4.2 Full-model optimality and selection

Theorem 2 (Complete convex KKT certificate). *Under a convex constraint qualification, or for the bounded finite-scenario linear programme, a feasible solution of Eq. 4 is optimal if and only if there exist nonnegative multipliers for every inequality, together with the free-threshold and scenario-excess dual variables, that satisfy primal feasibility, dual feasibility, stationarity Eq. 8 and complementarity.*

The proof follows convex KKT sufficiency or, in the implemented model, linear programming strong duality. The theorem matters because incomplete stationarity can assign a residual margin difference to “risk” even when budget, contract or shared capacity is active. It also clarifies the role of multiple optima. KKT validates a point; optimal-face analysis determines whether the ranking relation holds throughout all equivalent optima.

4.3 Dependence paths

Proposition 1 (Named-family feasible-set ordering). *Fix all crop marginals. If $L_{\theta_1}(x) \preceq_{\text{cx}} L_{\theta_2}(x)$ for every x in a declared domain, then*

$$\text{CVaR}_{\alpha,\theta_1}[L(x)] \leq \text{CVaR}_{\alpha,\theta_2}[L(x)].$$

If the domain is all of $\mathcal{X}$, the risk-feasible set under θ_2 is contained in that under θ_1 , so optimal expected profit cannot increase. Individual crop shares need not be monotone.

CVaR is monotone in convex order, yielding the risk and feasible-set inclusions. But set inclusion does not order coordinates of different optimizers. We therefore report Gaussian, Student- t and Clayton results family by family, and we retain active-set transitions, multiple optima and disconnected regions.

4.4 An executable diversification-failure criterion

Let x^{MV} be the policy chosen under a conventional Gaussian mean–variance assessment and x^T the full loss-CVaR optimum under the declared tail law.

Definition 2 (Diversification failure). *Conventional diversification fails when*

$$\text{Var}_G[\Pi(x^{MV})] < \text{Var}_G[\Pi(x^0)], \quad (9)$$

$$\text{CVaR}_{\alpha,T}[L(x^{MV})] > \text{CVaR}_{\alpha,T}[L(x^T)], \quad (10)$$

$$x^{MV} \neq x^T, \quad (11)$$

where x^0 is the expected-profit policy under the matched Gaussian law. In computation, allocation disagreement requires $\|x^{MV} - x^T\|_1 > \varepsilon_x$, and tail inferiority requires $\text{CVaR}_{\alpha,T}[L(x^{MV})] > \text{CVaR}_{\alpha,T}[L(x^T)] + \varepsilon_r$. Failure is operationally strong when $\text{CVaR}_{\alpha,T}[L(x^{MV})] > \kappa + \varepsilon_r$.

Proposition 2. *If x^T is feasible and the preceding three conditions hold, x^{MV} is not tail-risk optimal. Under the strong inequality it also violates the declared risk ceiling.*

This statement is logically modest but empirically useful. It separates variance reduction, matched Gaussian dependence, evaluation-law loss-CVaR and allocation disagreement. It does not assert that every diversified portfolio fails or that tail dependence can be summarized by one coefficient. If $\text{CVaR}_{\alpha,T}[L(x^T)] = \kappa$, the tail-inferiority and ceiling-violation inequalities coincide numerically and must not be counted as independent evidence.

4.5 Information, complementarity and substitution

Nature draws state ω , produces a noisy signal S , and leaves a fraction of the plan adjustable. The producer observes S , chooses a signal-contingent action and then realizes margin. Loss-CVaR is evaluated ex ante over the combined state–signal distribution.

Theorem 3 (Restricted information value). *In a posterior-separable model, or when the shared ex-ante risk restriction is nonbinding, if ignoring the signal is admissible, optimized information value is nonnegative. It is zero when one common action is optimal for every posterior. If posterior optima are unique and differ on positive-probability signal events, with strict improvement on at least one event, information value is strictly positive.*

The constant policy reproduces the no-information solution, proving nonnegativity. A common posterior optimum proves zero value. Strict posterior improvement integrates to positive value in the stated submodel. It does not establish strictness when signal-contingent actions share a binding ex-ante CVaR budget. Nested action sets raise both informed and uninformed optimized values, but their difference need not be monotone.

Proposition 3 (Restricted conditional complementarity). *In the posterior-separable or nonbinding-risk submodel, if actions are nested and state–action payoff has increasing differences in signal precision and acreage flexibility, and posterior-optimal actions differ on events of positive probability, information and flexibility are strictly complementary.*

Proposition 4 (Substitution through shock buffering). *Let flexibility ϕ contract state-dependent margins toward a common vector,*

$$\pi_\omega(\phi) = (1 - \phi)\pi_\omega + \phi\bar{\pi}.$$

At $\phi = 1$, the state no longer changes margin and information has zero value. If information value is positive at $\phi = 0$ and the value function is continuous, it must decrease on at least one interval; shock-buffering flexibility substitutes for information there.

These results do not support unconditional supermodularity. For the complete shared-ex-ante-CVaR numerical programme we instead compute, on every adjacent grid rectangle,

$$\Delta_{q,\phi}V = [V(q_2, \phi_2) - V(q_1, \phi_2)] - [V(q_2, \phi_1) - V(q_1, \phi_1)].$$

Positive, zero/boundary and negative cross-differences are numerical classifications, not an extension of the restricted theorem.

5 Public-data calibration and numerical experiments

5.1 Calibration

Kansas was selected because corn, soybean and winter wheat have complete public observations in every 2016–2024 analysis year and form a relevant diversified crop set. The 2016–2023 observations calibrate the score and margins; 2024 is excluded from the score and retained for external evaluation.

The score for crop i is the mean of Kansas yield divided by same-year national NASS yield over 2016–2023. It measures relative historical yield potential, not revenue. Scores are 0.748 for corn, 0.756 for soybean and 0.894 for winter wheat. Real per-acre margin combines Kansas yield, national harvest price, national operating cost and CPI-U conversion to 2024 dollars. Mean margins are \$ 199.8, \$ 229.0 and \$ 138.3, respectively. These aggregate inputs are transparent calibration values, not farm-level profits.

The primary marginal law is Student- t with five degrees of freedom. The primary joint law is a Student- t copula with four degrees of freedom and Kendall’s $\tau = 0.25$, simulated with 512 optimization scenarios. Sensitivity uses Gaussian and empirical-quantile margins; Gaussian, Student- t and Clayton copulas; and $\tau \in \{0, 0.15, 0.30, 0.45, 0.60\}$. The eight annual observations cannot estimate extreme tails precisely, so these are structural stress paths rather than fitted farm copulas.

Land is normalized to one. Crop lower bounds are 0.05, 0.10 and 0.05; upper bounds are 0.55, 0.65 and 0.60. The operating budget is \$330, corn is capped at 0.50 by rotation, soybean has a 0.10 contract minimum, and planting labour and harvest equipment have shared crop-specific coefficients. The primary confidence level is 0.95 and risk tolerance $\rho = 0.5$. All counterfactuals use the same full constraint system unless a constraint is the sensitivity dimension.

5.2 Primary reversal

The expected-profit endpoint without loss-CVaR assigns 0.280 to corn, 0.650 to soybean and 0.070 to winter wheat. It produces a complete rank reversal, but not a strong reversal because winter wheat remains positively allocated; its loss-CVaR \$ -34.5 , exceeding the primary risk ceiling. The minimum-CVaR endpoint attains \$ -59.5 but sacrifices expected profit. The midpoint ceiling is \$ -47.0 .

The full optimum assigns 0.330 to corn, 0.422 to soybean and 0.248 to winter wheat. The highest-ranked crop therefore receives the least land, a selected complete rank reversal involving both pairs, but not a strong exclusion reversal. The land and CVaR restrictions bind; idle land is zero. Expected profit is \$ 198.5 per normalized acre and realized finite-scenario loss-CVaR equals the ceiling to numerical precision. Maximum primal, stationarity and complementarity residuals are $1.42e - 14$, $2.84e - 14$ and $5.06e - 14$, respectively.

Ranking reversal occupies a family- and risk-dependent phase

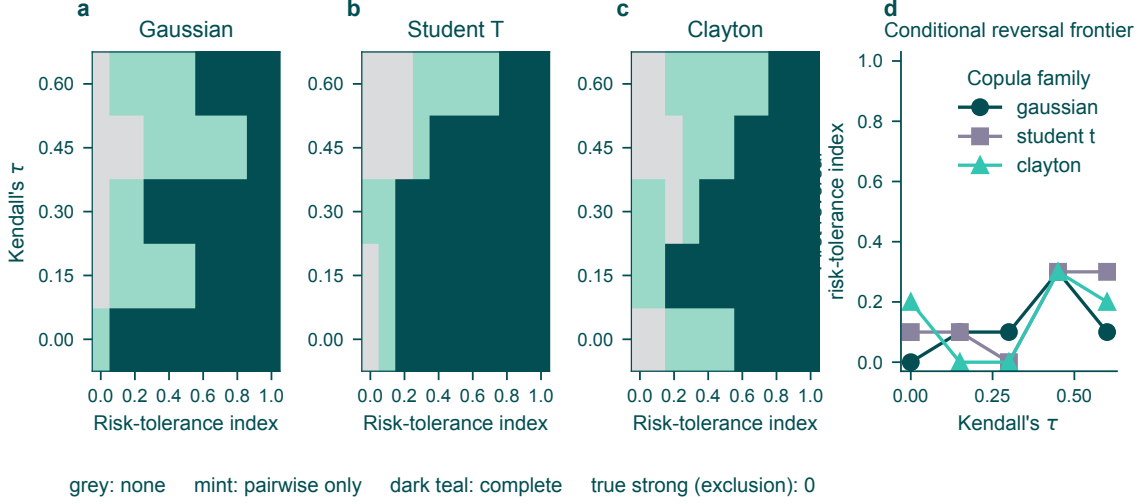


Figure 2: Ranking reversal occupies a family- and risk-dependent phase. a–c, Classification across risk tolerance and Kendall's τ for Gaussian, Student- t and Clayton copulas. Grey indicates no reversal, mint indicates pairwise-only reversal and dark teal indicates complete rank reversal; strong exclusion counts are stated separately. d, The first selected-reversal risk tolerance varies non-monotonically across the named family paths. All 165 evaluated cells, including 2 multiple-optimum cells, are retained.

This Kansas result is margin-induced: the score–margin separation favours soybean economically even though winter wheat ranks agronomically first, and the expected-profit endpoint is already inverted. The CVaR ceiling raises winter wheat from 0.070 to 0.248 and therefore moderates rather than creates the top-crop inversion. Operational constraints delimit the available path between those endpoints.

5.3 Conditional phase diagram

Figure 2 reports all 165 combinations of copula family, Kendall's τ and risk tolerance. Selected pairwise reversal occurs in 143 cells, selected complete rank reversal in 96 cells and strong exclusion reversal in 0 cells. There are no infeasible cells, but two Clayton cells have multiple optima. Most reversal regions are connected on the evaluated grid; Clayton at $\tau = 0.30$ is disconnected. The first selected-reversal index ranges from 0 to 0.3 across family and dependence paths. No strong-reversal index exists on any evaluated path because no cell satisfies the restored exclusion definition.

The nonmonotonic cells are substantively important. Increasing dependence can change which resource or risk constraint binds, and that active-set transition can move individual crop shares in either direction. The phase diagram therefore supports Proposition 1's feasible-set logic without claiming coordinate monotonicity. Scalar lower-tail dependence is reported as a diagnostic within each family, not used as a cross-family ordering statistic.

5.4 Diversification failure under matched rank dependence

The comparison x^0 is the matched-Gaussian expected-profit allocation. The mean–variance benchmark and tail-aware model use the same marginal means and standard deviations and matched Kendall's $\tau = 0.25$. The Student- t law nevertheless has lower-tail coefficient 0.195, whereas the Gaussian copula has zero asymptotic tail dependence. We solve a complete 301-point Gaussian mean–variance frontier over $\gamma \in [0, 0.03]$, with increments of 10^{-4} . The selected $\gamma = 0.0082$ is the smallest value attaining the fixed, outcome-independent target of at least 15% variance reduction relative to x^0 .

At the selected point, Gaussian variance falls from 6765.6 to 5733.5 (15.3%), while the allocation

Variance falls 15.3%, but the selected Gaussian policy is tail-inferior

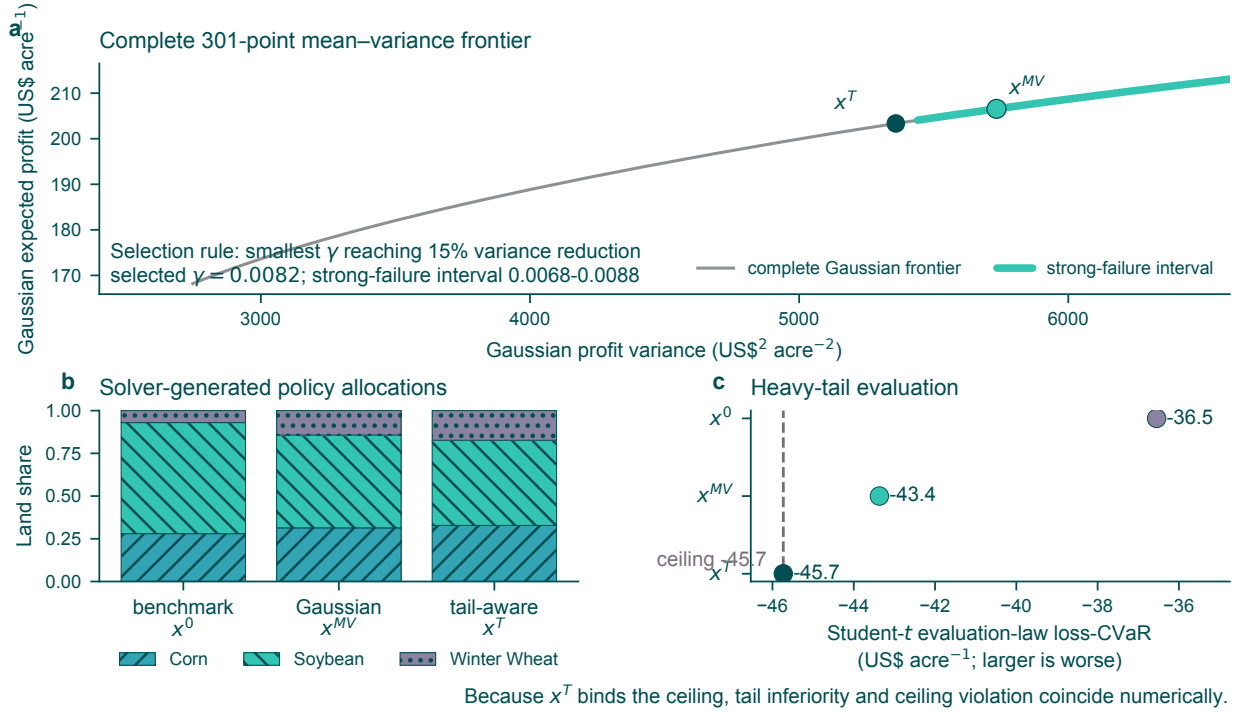


Figure 3: Variance diversification can fail in the joint lower tail. **a**, Complete Gaussian mean–variance frontier, the fixed variance reduction target and the resulting failure interval. **b**, Solver-derived allocations x^0 , x^{MV} and x^T . **c**, Student- t evaluation loss-CVaR for the same policies under a common ceiling. The tail-inferiority and ceiling-violation gaps coincide numerically because x^T binds that ceiling. Gaussian and Student- t laws match Kendall’s τ , but only the latter has positive asymptotic lower-tail dependence.

changes by 0.090 in L^1 distance. Under the Student- t evaluation law, the Gaussian mean–variance policy has loss-CVaR \$ -43.4 , exceeding the common ceiling \$ -45.7 by \$ 2.364 . The tail-aware policy changes the allocation and lowers evaluation loss-CVaR by \$ 2.364 . The weak and strong failure criteria hold over γ intervals 0.0068 – 0.0088 and 0.0068 – 0.0088, respectively.

Because the tail-aware optimum binds the common CVaR ceiling, its tail improvement and the Gaussian policy’s ceiling violation are numerically equal at the selected point; these are therefore two expressions of one tail-risk gap, not independent confirmations. Across 18 one-factor perturbations of target variance reduction, scenario count, seed, dependence, confidence level, risk tolerance and marginal family, 17 satisfy the strong criterion. The result identifies a specified region in which Gaussian variance diversification does not protect the joint lower tail; it is not a universal claim.

5.5 Mechanism isolation and policy robustness

The suitability-proportional, winner-take-all and equal-share policies are projected to operational feasibility before evaluation. Suitability proportional earns \$187.4 with loss-CVaR $-\$53.1$ and does not reverse. Feasibility-projected winner-take-all earns \$166.6 with loss-CVaR $-\$57.3$ and reverses one pair. Equal share earns \$190.4 with loss-CVaR $-\$51.4$. The expected-profit and mean–variance endpoints both earn \$216.5 but violate the CVaR ceiling. The full model earns less expected profit than those infeasible endpoints because it buys certified downside protection.

The controlled risk experiment uses the Kansas Soybean–Corn score and margin ordering: soybean has both the higher score and a mean-margin advantage of \$29.2. A mean-preserving 10% severe soybean

Margin, downside risk and operations leave distinct reversal signatures

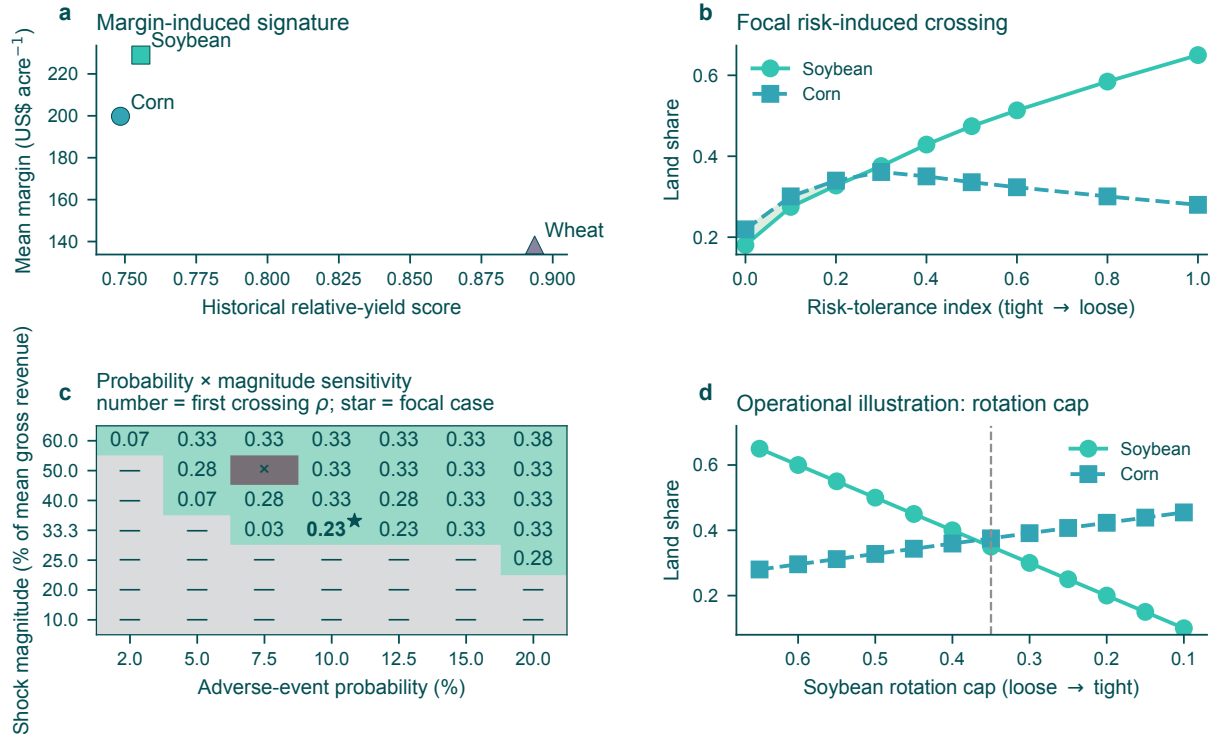


Figure 4: Three reversal mechanisms have distinct counterfactual signatures. **a**, Kansas score and mean-margin ranks separate, so the primary top-crop inversion is margin-induced. **b**, With soybean higher in both score and mean margin, tightening the shared loss-CVaR ceiling under a mean-preserving downside stress crosses the Soybean–Corn allocation order. **c**, Probability-by-magnitude sensitivity distinguishes crossing, non-crossing and infeasible stress cells. **d**, With risk nonbinding, tightening only the soybean rotation cap produces an operational crossing; all preceding null constraint stages remain in the source data.

drought-or-basis event shifts \$154.7 per acre into the downside state and compensates non-adverse states exactly. At loose risk tolerance, soybean and corn receive 0.650 and 0.280; under the tight ceiling they receive 0.181 and 0.219. This is a genuine risk-induced pairwise crossing with score and mean order held fixed. A probability-by-magnitude sensitivity map evaluates 49 stress combinations: 24 cross, 24 remain feasible without crossing and 1 are infeasible. In the focal cell, the first crossing occurs at risk tolerance 0.225. These controlled stresses isolate a mechanism; they do not estimate the probability or severity of a Kansas farm shock.

The operational experiment holds the score, margin scenarios and a deliberately nonbinding risk ceiling fixed. Adding the observed Kansas constraint sequence changes active sets but does not by itself reverse the Soybean–Corn pair. A controlled soybean rotation-cap path crosses the ranking boundary first at cap 0.35. Null stages are retained; this path is a structural mechanism, not an estimate of a private Kansas constraint (Fig. 4).

Every one-at-a-time sensitivity dimension retains selected pairwise reversal: α , constraint regime, dependence family, dependence parameter, marginal family, sample window, scenario count, seed and solver. Strong exclusion does not occur because crop lower bounds are positive. The solver comparisons use HiGHS, dual simplex and interior point; scenario counts are 256, 512 and 1,024; sample windows begin in 2016, 2017 or 2018. These robustness runs assess structural stability, not sampling-independent replication.

Information remains valuable, but its interaction with flexibility changes sign

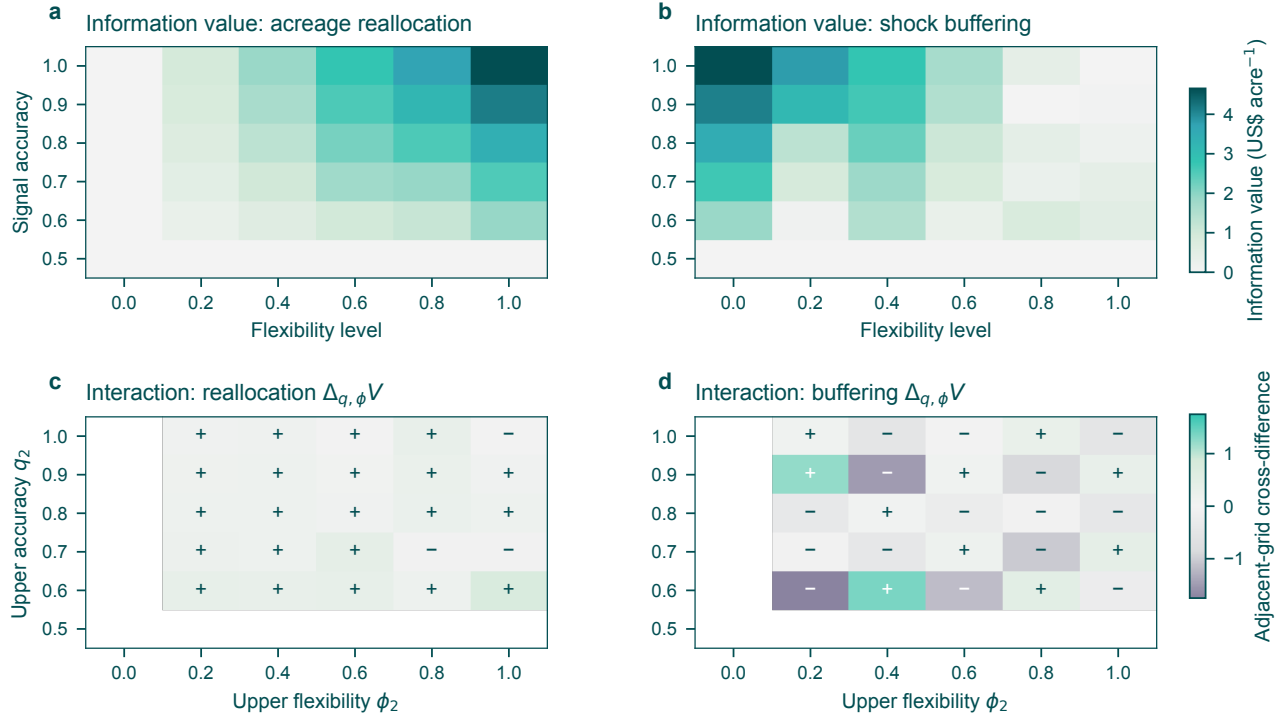


Figure 5: Information–flexibility interaction is classified on the solved value grid. a,b, Information value under post-signal acreage reallocation and shock buffering. **c,d,** Adjacent-grid cross-differences for the same paths, computed as $[V(q_2, \phi_2) - V(q_1, \phi_2)] - [V(q_2, \phi_1) - V(q_1, \phi_1)]$. Values are optimized under one shared ex-ante loss-CVaR ceiling.

5.6 Information and flexibility

The state proxy is whether the annual Kansas corn-minus-soybean relative-yield advantage lies below or above its 2016–2023 median. A noisy pre-plant signal has accuracy 0.5–1.0. The two-stage loss-CVaR restriction is imposed once on the ex-ante distribution across state, signal and contingent actions; it is not replaced by a separate risk budget for each signal.

With post-signal acreage reallocation, no adjustable share or an uninformative signal gives zero value. As accuracy and the adjustable share rise, posterior allocations diverge and information value reaches \$4.655 per acre. With shock buffering, more flexibility instead contracts state-specific margins toward a common vector. At perfect signal accuracy, information value declines from approximately \$4.4 without buffering to numerical zero under full buffering. The adjacent grid contains 32 positive, 17 negative, 18 zero-information and 5 zero/boundary classifications. These are discrete cross-difference results for the coupled model, not applications of the restricted complementarity theorem.

5.7 Uncertainty

Sixty-four historical resamples re-estimate scores and margins, regenerate scenarios and resolve the full optimum. Selected pairwise reversal occurs in 62 of 64 resamples, a frequency of 0.969. The exact Clopper–Pearson 95% binomial interval for that resample frequency is $[0.892, 0.996]$; it is not a bootstrap percentile interval or a farm-level prevalence interval. The first-reversal risk-tolerance index averages 0.066 with interval $[0.000, 0.598]$, revealing substantial uncertainty in the boundary even though reversal itself is common. Mean bootstrap allocations are 0.277 for corn, 0.527 for soybean and 0.196 for winter wheat;

their percentile intervals are wide because only eight years enter each calibration. This is the appropriate limitation to emphasize: the qualitative reversal is stable, while its exact frontier and crop shares remain imprecise.

6 External descriptive evidence

6.1 Purpose and construction

The external analysis asks whether score–acreage disagreement is visible beyond the Kansas calibration. It does not treat observed state acreage as a solution to Eq. 4. USDA NASS annual state reports yield 248 complete state-years across 31 states from 2016 through 2024 for corn, soybean and winter wheat. National harvest prices and operating and total costs come from USDA ERS, and BLS CPI-U supplies the real-dollar conversion. Suppressed or missing acreage and yield are not imputed.

Four ranking definitions are computed before comparison with planted-acreage share. Relative yield is state yield divided by same-year national yield. Standardized revenue multiplies state yield by national price. Operating margin subtracts national operating cost; total-cost margin subtracts total cost. The latter three are transparent accounting measures rather than local or private expected profits. For each state-year we calculate pairwise inversion intensity and whether the top-ranked crop differs from the acreage leader. State-cluster bootstrap intervals preserve within-state serial dependence.

6.2 Descriptive prevalence and definition sensitivity

Top-rank disagreement is 0.887 for relative yield, 0.613 for standardized revenue, 0.625 for operating margin and 0.782 for total-cost margin. The corresponding state-cluster intervals are all well above zero. The ordering across definitions is itself a result: a claim about ranking reversal inherits the scientific meaning of the score being ranked. Relative-yield disagreement cannot be interpreted as profit inefficiency, while a margin-based disagreement remains incomplete because local basis, heterogeneous costs, contracts, beliefs and risk limits are unobserved.

The leakage-free temporal check uses prior relative-yield potential to predict the following change in acreage share with crop, state and decision-year fixed effects. It includes 651 crop-transition rows. The estimated effect is -0.0042 share units with interval $[-0.0123, 0.0056]$, which includes zero. Thus the panel does not show that a crop’s earlier score leadership leads to a larger next-year share change. This null is consistent with many mechanisms, including persistent rotations, adjustment cost and a score that does not measure private profit; it is not evidence that agronomic information has no value.

6.3 Evidence boundary

The external layer validates three limited propositions. Score–acreage disagreement exists at a meaningful scale; it varies across definitions; and aggregate data do not show a leakage-free lagged response to the genuine relative-yield score. It does not identify the full model’s CVaR mechanism, estimate private risk aversion, reveal the active constraint set, or establish welfare. Those claims require farm-level expectations, resource capacities, contract terms and decision timing. Treating the state panel as descriptive is not a weakness to hide; it prevents aggregate acreage from being used to “confirm” a mechanism that it cannot identify.

7 Managerial and policy implications

7.1 Recommendations should expose their decision layer

A crop-ranking tool can remain useful without prescribing acreage. The problem arises when an ordinal output is presented as if the top crop should receive the most land. A prescriptive system should expose

External disagreement is descriptive and sensitive to definition and record length

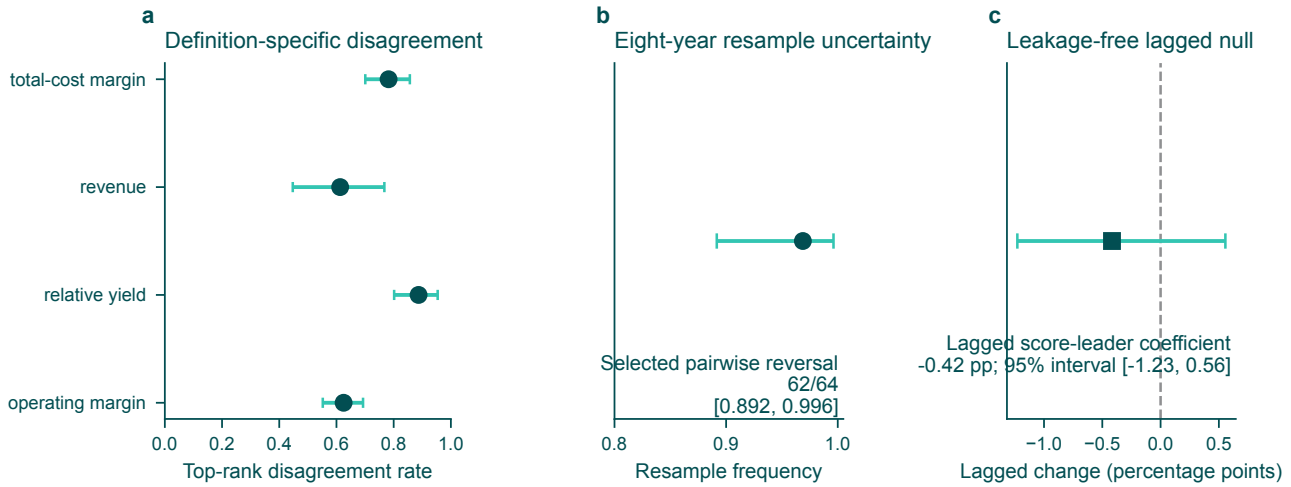


Figure 6: Resampling quantifies short-record sensitivity, not farm-level prevalence. **a**, Top-rank disagreement rates and state-cluster 95% intervals in 248 state-years. **b**, Selected pairwise reversal occurs in 62 of 64 historical resamples; the bar is the exact Clopper–Pearson 95% binomial interval. **c**, Leakage-free lagged association between prior score leadership and next-year acreage-share change.

at least five objects: the cardinal margin model, the joint distribution used to form portfolio losses, the operational feasible set, the risk convention and the selector used when optima are multiple. If these objects are unavailable, the defensible output is a rank or scenario comparison, not a claim of optimal acreage.

The distinction matters for accountability. A lender or platform that flags a farmer for planting less of a high-scoring crop may be penalizing compliance with a rotation, contract or liquidity restriction omitted from the score. Conversely, saying that every disagreement is rational would make the recommendation system unfalsifiable. The remedy is to compare declared feasible counterfactuals and characterize the optimal face.

7.2 Downside-risk governance

The Kansas experiment shows the cost of treating a mean–variance solution as tail safe. Under matched rank dependence, the conventional policy earns a higher mean but violates the Student-*t* evaluation-law CVaR ceiling. This suggests a governance test for decision-support systems: report the allocation under at least one heavy-tailed joint stress law, evaluate it under the same downside-risk metric used for the farmer’s tolerance, and disclose whether the policy remains feasible. The test is more informative than reporting a diversification index or pairwise correlation alone.

The result does not justify universal conservatism. A very tight risk ceiling reduces expected profit and can alter which crop dominates; a loose ceiling may leave the expected-profit allocation unchanged. Decision makers should therefore report the entire risk frontier and active-set transitions. The location of reversal is conditional on that frontier, not an immutable property of a crop.

7.3 Invest in information only with an action pathway

Forecast accuracy and operational flexibility should be evaluated jointly. When a share of acreage can be adjusted after a credible pre-plant signal, better information can be increasingly valuable. If planting is

already fixed or posterior-optimal plans coincide, more signal accuracy has zero economic value even if the prediction is statistically informative. These are design criteria for climate services: document decision timing and the specific acreage, input, irrigation or harvest actions that remain open.

Shock-buffering flexibility creates a different investment trade-off. Irrigation, robust varieties, input switching or flexible harvest capacity can reduce exposure to the predicted state. This may raise total profit while reducing the incremental value of a forecast. Substitution is not a failure of the forecast; it means the operational technology has made the distinction between states less consequential. A portfolio of information and flexibility should therefore be assessed by total optimized value, not by requiring the interaction to be positive.

7.4 Policy interpretation

Rotation and diversification mandates should distinguish crop count from joint-tail protection. A policy that increases the number of planted crops can still leave a producer exposed to simultaneous drought losses when the crops share phenological vulnerabilities [Schmitt et al., 2024]. Crop-pair choice, timing, geography and access to off-farm risk instruments may matter more than a generic diversity threshold.

Likewise, public rankings based on climate suitability should not be evaluated solely against observed acreage. State acreage aggregates many farms and combines agronomic, financial and institutional decisions. The appropriate policy question is whether the ranking improves a declared decision under realistic constraints. The external panel in this study motivates that question but cannot answer it causally.

8 Conclusion

This study returns crop-ranking reversal to its intended scientific foundation: a conditional allocation phenomenon generated by stochastic margins, joint downside risk and operational feasibility. A genuine agronomic score remains separate from profit. A full loss-CVaR-constrained programme then determines whether the high-ranked crop receives more or less land.

The restricted two-crop theorem provides an exact frontier rather than an informal risk-adjusted ranking. When the higher-ranked crop also has the higher mean margin, reversal occurs precisely when the risk-feasible interval ends below the equal-share boundary. Unique dependence thresholds require named-family monotonicity and crossing; otherwise reversal regions can be nonmonotone or disconnected. The complete KKT system and optimal-face bounds prevent local multipliers or one solver vertex from being overinterpreted.

In the public-data-calibrated Kansas experiment, winter wheat ranks first in historical relative-yield potential but receives the least land in the full model. This is a selected complete rank reversal, not a strong exclusion reversal, and it is primarily margin-induced; CVaR moderates it. Controlled experiments separately exhibit risk- and operationally induced Soybean–Corn crossings. A full-investment matched-Kendall Gaussian mean–variance point selected by a fixed 15% variance-reduction target strictly reduces variance relative to its expected-profit benchmark yet violates the Student- t loss-CVaR ceiling. The failure persists over a non-singleton γ interval and is parameter-dependent across one-factor sensitivity cases. Information–flexibility interaction in the shared-CVaR model is reported from signed adjacent-grid cross-differences; the strict theorem remains restricted to a separable or nonbinding-risk submodel.

The 248-state-year panel shows that score and acreage leaders often differ, but its leakage-free lagged estimate includes zero. Aggregate prevalence therefore cannot identify private risk, constraints or welfare. Future empirical work needs farm-level pre-plant beliefs, local prices and costs, crop histories, contracts, credit, capacity and signal receipt.

The final message is operational. Rankings can inform planting, but only a transparent decision model can prescribe acreage. That model should report its joint tail law, risk frontier, active constraints, multiple-optimum classification and the action pathway through which information can matter.

9 Methods

9.1 Data sources and transformations

State acreage and yield are obtained from USDA NASS annual reports [U.S. Department of Agriculture, National Agricultural Statistics Service, 2019, 2022, 2025]. National crop yield and harvest price are from USDA NASS, and operating and total costs are from USDA ERS [U.S. Department of Agriculture, Economic Research Service, 2026]. Annual CPI-U from the US Bureau of Labor Statistics converts monetary values to 2024 dollars [U.S. Bureau of Labor Statistics, 2026]. Crop-year panels are formed by complete-case joins; no missing or suppressed agricultural value is imputed.

The Kansas score uses 2016–2023. For each crop and year, state yield is divided by the national yield of the same crop; the score is the eight-year mean. Margin is state yield times national harvest price minus national operating cost, then expressed in real 2024 dollars. Student- t , Gaussian and empirical-quantile marginals are calibrated to the historical means and dispersion. The primary score intentionally excludes the 2024 evaluation year.

9.2 Scenario generation and optimization

Common random numbers are used within each model contrast. Primary optimization uses 512 equally weighted scenarios and seed 202607270034. Copula correlation is derived from matched Kendall’s τ . The Student- t copula uses four degrees of freedom and Student- t marginals use five. Linear programmes are solved with HiGHS. Every solution is re-evaluated from scenario profit, resource use and constraint violation, and its primal–dual conditions are checked.

Minimum-CVaR and expected-profit endpoints are solved separately for each joint law. Equation 7 produces eleven risk ceilings. To characterize objective-equivalent allocations, expected profit is constrained to its optimum within numerical tolerance and each ranked-pair difference is minimized and maximized. Complete-rank and exclusion feasibility are checked on the same optimal face. The deterministic HiGHS allocation defines the selected classification; universal and possible classifications use the face bounds. Ordering and zero tolerances are 10^{-4} normalized land share, with sensitivity from 10^{-8} to 10^{-2} .

9.3 Diversification comparison

The benchmark x^0 maximizes expected profit under the matched Gaussian model. For each $\gamma \in \{0, 10^{-4}, \dots, 0.03\}$, $x^{MV}(\gamma)$ maximizes Gaussian mean minus γ times Gaussian variance, subject to the same full-investment and operational constraints. The reported mean–variance policy is selected without reference to its tail outcome: it is the smallest γ that reduces Gaussian variance by at least 15% relative to x^0 . Targets of 10% and 20% are sensitivity cases.

The tail-aware policy x^T maximizes expected profit subject to a common Student- t loss-CVaR ceiling. All three allocations are then evaluated under that Student- t law. Weak failure requires the fixed Gaussian variance target, a nonzero allocation difference and higher evaluation loss-CVaR for x^{MV} than for x^T . Strong failure additionally requires x^{MV} to violate the common ceiling. Sensitivity varies target, scenario count, seed, dependence strength, copula degrees of freedom, confidence level, risk tolerance and marginal family one factor at a time.

9.4 Mechanism isolation and robustness

Score-proportional, score-winner and equal-share allocations are projected onto the operationally feasible set before outcome comparison. Expected-profit, mean–variance, full loss-CVaR and minimum-CVaR endpoints use the same constraints. Robustness varies one dimension at a time: confidence level, dependence family and parameter, marginal model, historical window, scenario count, seed, solver and removal of each constraint group. Null, nonmonotone, multiple-optimum and infeasible cases are retained.

The risk-isolation path preserves the Kansas Soybean–Corn score and mean orders, then applies a mean-preserving severe soybean downside event. Probability ranges from 2% to 20% and adverse gross-revenue share from 10% to 60%; risk tolerance spans 0 to 1. The operational path fixes margin scenarios and a nonbinding risk ceiling, adds observed-resource constraint groups sequentially and then tightens only the soybean rotation cap. These controlled stresses identify mathematical mechanisms rather than estimating private farm constraints or shock probabilities.

Historical uncertainty resamples calibration years 64 times. Each replication re-estimates scores and margins, regenerates scenarios and resolves the frontier. Continuous outputs use historical-resample percentile intervals. Binary reversal frequencies use exact Clopper–Pearson binomial intervals. Neither interval treats simulated scenarios as independent empirical observations.

9.5 Information and flexibility

The binary state is constructed from the corn-minus-soybean relative-yield advantage around its historical median. Signal accuracy ranges from 0.5 to 1.0. A feasible constant policy is always available, providing an exact signal-ignorability check. The acreage-recourse path changes the share adjustable after signal arrival. The buffering path contracts state-dependent margins toward their common mean while retaining signal-contingent actions. One loss-CVaR ceiling applies to the full ex-ante state–signal loss distribution. Numerical information values below 10^{-7} are classified as zero. For each adjacent $q_1 < q_2$, $\phi_1 < \phi_2$ rectangle, interaction is the solved cross-difference $[V(q_2, \phi_2) - V(q_1, \phi_2)] - [V(q_2, \phi_1) - V(q_1, \phi_1)]$.

9.6 External descriptive analysis

For the 31-state panel, pairwise inversion intensity counts discordant pairs among the three crops. Top-rank disagreement equals one when the highest-score crop is not the largest-acreage crop. Confidence intervals use state-cluster resampling. The lagged model predicts next-year acreage-share change from the previous relative-yield leader and previous acreage share, with crop, state and decision-year fixed effects. It uses no contemporaneous score to predict the same year’s acreage.

9.7 Data availability

The public source reports, analysis-ready tables and figure source data needed to reproduce the reported results will be deposited in a versioned public archive upon publication. No confidential farm-level data are used.

9.8 Code availability

Analysis code sufficient to regenerate the numerical results and figures will be deposited with the source data upon publication. Additional mathematical proofs, phase classifications, data dictionaries and numerical diagnostics are provided in Supplementary Information.

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Author contributions

X.L. conducted the data analysis, developed the computational experiments and drafted the manuscript. P.C. conceived and supervised the study. Both authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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